Band-1 Hadamard order 13904 – Classical Paley-I (q=13903); IN SPK via Paley-I; Gap 2052 Still OPEN

H_13904 Paley-I PASS; Paley-II FAIL/SKIP (q=6951=3*7331 NOT prime; ===3 mod 4 needs ===1); Kronecker SKIP note-only (*4 3476; 3476===0 mod 4 order-admissible; factors 2^2*11*79; not_in_seeds / no H_3476; H_3476 has NO direct classical Paley route (Paley-I q=3475=5^2*139* composite; Paley-II q=1737=3^2*193 composite); 3476=2*1738 (1738===2 mod 4 NOT Hadamard-admissible – no Sylvester-doubling route) – note-only, NOT constructed, not on box); Miyamoto digression skipped; gap 2052 OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-10-07

Document class. Scientific computational / construction certificate (Band-1).

Program. Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.

Parents. 2026-10-07_Band1_Hadamard_13884_PaleyI; 2026-10-07_Band1_Hadamard_13880_PaleyI; 2026-10-07_Band1_Hadamard_13860_PaleyI; 2026-10-07_Band1_Hadamard_13832_PaleyI; 2026-10-07_Band1_Hadamard_13808_PaleyI; 2026-10-07_Band1_Hadamard_13800_PaleyI; 2026-10-07_Band1_Hadamard_13764_PaleyI; 2026-10-07_Band1_Hadamard_13760_PaleyI; 2026-10-07_Band1_Hadamard_13752_PaleyI; 2026-10-07_Band1_Hadamard_13724_PaleyI; 2026-10-07_Band1_Hadamard_13712_PaleyI; 2026-10-07_Band1_Hadamard_13692_PaleyI; 2026-10-07_Band1_Hadamard_13688_PaleyI; 2026-10-06_Band1_Hadamard_13680_PaleyI; 2026-10-06_Band1_Hadamard_13628_PaleyI; 2026-10-06_Band1_Hadamard_13620_PaleyI; 2026-10-06_Band1_Hadamard_13592_PaleyI; 2026-10-06_Band1_Hadamard_13568_PaleyI; 2026-10-06_Band1_Hadamard_13524_PaleyI; 2026-10-06_Band1_Hadamard_13500_PaleyI; 2026-10-06_Band1_Hadamard_13488_PaleyI; 2026-10-06_Band1_Hadamard_13464_PaleyI; 2026-10-06_Band1_Hadamard_13452_PaleyI; 2026-10-06_Band1_Hadamard_13412_PaleyI; 2026-10-06_Band1_Hadamard_13400_PaleyI; 2026-10-06_Band1_Hadamard_13368_PaleyI; 2026-10-06_Band1_Hadamard_13340_PaleyI; 2026-10-06_Band1_Hadamard_13332_PaleyI; 2026-10-06_Band1_Hadamard_13328_PaleyI; 2026-10-06_Band1_Hadamard_13292_PaleyI; 2026-10-06_Band1_Hadamard_13268_PaleyI; 2026-10-06_Band1_Hadamard_13260_PaleyI; 2026-10-06_Band1_Hadamard_13220_PaleyI; 2026-10-06_Band1_Hadamard_13188_PaleyI; 2026-10-06_Band1_Hadamard_13184_PaleyI; 2026-10-06_Band1_Hadamard_13172_PaleyI; 2026-10-05_Band1_Hadamard_13164_PaleyI; 2026-10-05_Band1_Hadamard_13160_PaleyI; 2026-10-05_Band1_Hadamard_13152_PaleyI; 2026-10-05_Band1_Hadamard_13148_PaleyI; 2026-10-05_Band1_Hadamard_13128_PaleyI; 2026-10-05_Band1_Hadamard_13104_PaleyI; 2026-10-05_Band1_Hadamard_13100_PaleyI; 2026-10-05_Band1_Hadamard_13064_PaleyI;

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2026-09-23_Band1_Hadamard_10100_PaleyI; 2026-09-23_Band1_Hadamard_10068_PaleyI.

Lab. `/workspace/experiments/band1/hadamard_13904_paleyI/`.

Cutoff. 2026-10-07T22:33:05Z (~6:33PM ET).

Honesty stamp (non-negotiable).

This memo claims **finite construction of order 13904 only (verified max|HHˆT-nI|=0), via classical Paley-I q=13903 (prime ===3 mod 4). 13904 IN SPK via Paley-I.** Honest skip: Miyamoto digression this wave (prefer classical SPK). Classical SPK for 13904: Paley-I **PASS** ($q = 13903$ prime $\equiv 3 \pmod 4$); Paley-II **FAIL/SKIP** (q=6951=3 7331 NOT prime; ===3 mod 4 needs ===1); Kronecker **SKIP note-only** (note $13904 = 4 \cdot 3476$; $3476 \equiv 0 \pmod 4$ order-admissible; factors $2^2 \cdot 11 \cdot 79$; not_in_seeds / no H_3476; H_3476 has NO direct classical Paley route

(Paley-I q=3475=5^2*139 composite; Paley-II q=1737=3^2193 composite); 3476=2*1738 (1738===2 mod 4 NOT Hadamard-admissible – no Sylvester-doubling route) – note-only, NOT constructed, not on box). **13904** folds into SPK **only** via classical Paley-I (this run). It **does not** claim the full Hadamard conjecture. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims. Gap **2052** remains **OPEN**. `clay_closed = false`. Prior SPK **13884+13880+13860+13832+13808+13800+13764+13760+13752+13724+13712+13692+13688+136** retained; Miyamoto-only **8948/8644/8516/8452/7988/...** excluded.

ewpage

Abstract

Finite construction of order **13904**.

- H_{13904} = Paley-I ($q = 13903$) – classical SPK
- Paley-II for 13904: **FAIL/SKIP** ($q = 6951 = 3 \cdot 7 \cdot 331$ NOT prime; $\equiv 3 \pmod 4$ needs $\equiv 1$)
- Kronecker for 13904: **SKIP note-only** ($4 \cdot 3476$; $3476 \equiv 0 \pmod 4$ order-admissible; factors $2^2 \cdot 11 \cdot 79$; not_in_seeds / no H_3476; H_3476 has NO direct classical Paley route (Paley-I q=3475=5^2*139 composite; Paley-II q=1737=3^2193 composite); 3476=2*1738 (1738===2 mod 4 NOT Hadamard-admissible – no Sylvester-doubling route) – note-only, NOT constructed, not on box)
- Skip: Miyamoto digression this wave

SPK: IN SPK via Paley-I.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_13904.py`

Results. overall_pass=True; units 621/621; row blocks 109/109 (block size 128); wall 2596.1474s; constructed max|HH^T-nI|=0.

Cutoff: 2026-10-07T22:33:05Z (~6:33PM ET)

Arithmetic

- Target $n = 13904 = q + 1$ with $q = 13903$ prime $\equiv 3 \pmod 4$ — Paley-I eligible.
- Paley-II: $q = 6951 = 3 \cdot 7 \cdot 331$ NOT prime; $\equiv 3 \pmod 4$ needs $\equiv 1$ — FAIL/SKIP.
- Kronecker: $13904 = 4 \cdot 3476$; $3476 \equiv 0 \pmod 4$ order-admissible; factors $2^2 \cdot 11 \cdot 79$; not_in_seeds / no H_3476; H_3476 has NO direct classical Paley route (Paley-I q=3475=5^2*139 composite; Paley-II q=1737=3^2193 composite); 3476=2*1738 (1738===2 mod 4 NOT Hadamard-admissible – no Sylvester-doubling route) – note-only, NOT constructed, not on box — SKIP note-only.
- Miyamoto digression skipped this wave (prefer classical SPK).

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_13904 Paley-I	Paley-I q=13903	0	2557.6806
H_13904 Paley-II	FAIL/SKIP (q=6951=3*7*331 NOT prime; ===3 mod 4 needs ===1)	–	–
H_13904 Kronecker	SKIP note-only (4*3476; 3476===0 mod 4 order-admissible; factors 2ˆ2*11*79; not_in_seeds / no H_3476; H_3476 has NO direct classical Paley route (Paley-I q=3475=5ˆ2*139 composite; Paley-II q=1737=3ˆ2*193 composite); 3476=2*1738 (1738===2 mod 4 NOT Hadamard-admissible – no Sylvester-doubling route) – note-only, NOT constructed, not on box)	–	–

SPK closure note

Seeds include prior SPK through **11528** lineage (**11528**; **11520**; **11504**; **11492**; **11484**; **11472**; **11468**; **11448**; **11444**; **11424**; **11412**; **11400**; **11384**; **11352**; **11312**; **11300**; **11288**; **11280**;

11252; **11244**; **11240**; **11172**; **11160**; **11132**; **11120**; **11088**; **11084**; **11072**; **11060**; **11048**; **11028**; **11004**; **10988**; **10980**; **10940**; **10904**; **10892**; **10884**; **10860**; **10848**; **10832**; **10800**; **10772**; **10740**; **10724**; **10712**; **10692**; **10688**; **10668**; **10652**; **10640**; **10632**; **10628**; **10608**; **10568**; **10560**; **10532**; **10500**; **10488**; **10464**; **10460**; **10428**; **10400**; **10392**; **10344**; **10332**; **10304**; **10272**; **10268**; **10260**; **10248**; **10244**; **10224**; **10212**; **10164**; **10160**; **10152**; **10140**; **10112**; **10104**; **10100**; **10068**; **10040**; **10008**; **9968**; **9932**; **9924**; **9908**; **9888**; **9884**; **9872**; **9860**; **9852**; **9840**; **9812**; **9804**; **9792**; **9788**; **9768**; **9744**; **9740**; **9720**; **9680**; **9644**; **9632**; **9624**; **9620**; **9588**; **9552**; **9548**; **9540**; **9512**; **9492**; **9480**; **9468**; **9420**; **9404**; **9392**; **9372**; **9344**; **9324**; **9320**; **9312**; **9284**; **9240**; **9228**; **9204**; **9200**; **9188**; **9152**; **9104**; **9128**; **9124**; **9092**; **2220+8884**; **2236**; **2160**; **2140+8564**; . . . ; **NOT** 8948/8644/8516/. . .) + verified **10464** (+ prior **10392+10344+10332+10304+10272+10268+10224+10212+10164+10160+10152+10140+10112+101**
Next constructive gap **2052** (OPEN).

Quantity	Value
12488 in SPK	**True** (Paley-I q=13903)
Gap 2052	**OPEN**
clay_closed	false
GATE-04	LOCKED

Honesty

- clay_closed = false
- GATE-04 LOCKED
- Hadamard conjecture OPEN
- Gap 2052 OPEN (method triage / non-finding only)
- No RH / P!=NP / YM / Frankl / NS / AC solved claims
- SPK membership only via classical Paley-I/II/Kronecker (not Miyamoto alone)

Next optional

q=13907->n=13908 Paley-I DIRECT (prime ===3 mod 4; no prime at all in 13904..13906 (13905=3^3*5*103, 13906=2*17*409); Paley-II q=6953=17*409 *NOT prime (not a prime power); ===1 mod 4 but composite -> FAIL/SKIP; Kronecker 13908=4*3477*; 3477=3*1961 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3477); 13908=2^2*3*19*61 ===4 mod 16 = odd-m shape (q2 composite ===1, m odd ===1 mod 4; cf. 13860)) (or gap 2052 / Track B)